

Linear Algebra:

For linear algebra tasks, write your answers in physical papers in English, including the details of how you get the final answers.

Task 1:

Determine whether the following two vectors are parallel.

- $(-2, 1, 3)$ and $(4, 6, 1)$
- $(1, 2)$ and $(-3, -6)$
- $(1, -2, 0, 1)$ and $(3, 0, 2, -5)$
- $(10, 0, 2, -4, -8)$ and $(5, 0, 1, -2, -4)$

Task 2:

Find the equations of the planes containing the following points in space. If the provided points do not define a specific plane, justify your answer.

- $(2, -5, -1)$, $(0, 4, 6)$, and $(-3, 7, 1)$
- $(1, 2, 1)$, $(2, 4, 2)$, and $(-3, -6, -3)$
- $(1, 1, 1)$, $(2, 5, 2)$, and $(0, 0, 0)$

Equation format should be $x = a + su + tv$ where s, t are arbitrary real numbers, and x, a, u, t are vectors.

Task 3:

Answer the following questions.

- Write the zero vector of $M_{2 \times 5}(\mathfrak{R})$ where $\mathfrak{R}$ is the field of real numbers.
- We define $P(F)$ as the set of all polynomials with coefficients from field F .

If $f(x) = 3.2x^2 + 2x - 0.5$, justify that if f belong to $P(\mathfrak{N})$ where $\mathfrak{N}$ is the field of natural numbers.

Task 4:

Label the following statements as true or false, give a counterexample if you labeled false.

- If f and g are polynomials of degree n , then $f + g$ is polynomial of degree n .
- An $m \times n$ matrix has n columns and m rows.
- In any vector space V , if $x, y \in V$, then $ax = ay$ implies that $x = y$.

WeHelp

Assignment - Week 1

Python:

For Python tasks this week, write all code without third-party libraries in a single program where we can execute it and get all the expected outputs at once.

Task 1:

Parse the items of ASUS PCs which are shown on the following URL.

<https://24h.pchome.com.tw/store/DSAA31>

We should parse all the items shown on every page and we should assume we don't know how many pages are there in advance.

Create a file named products.txt and print product IDs, 1 per line.

Task 2:

Based on the data we parsed in Task 1. Create a file named best-products.txt and print product IDs with at least 1 review where average rating greater than 4.9

Task 3:

Based on the data we parsed in Task 1. Calculate the average price of ASUS PCs with Intel i5 processor. Just print it in the console.

Task 4:

We want to use z-score to standardize the prices of ASUS PCs where you can treat parsed data in Task 1 as statistical population.

Create a file named standardization.csv and print data as the following format.

```
ProductID,Price,PriceZScore
ProductID,Price,PriceZScore
...
```